Structures

Group 8

March 16, 2026

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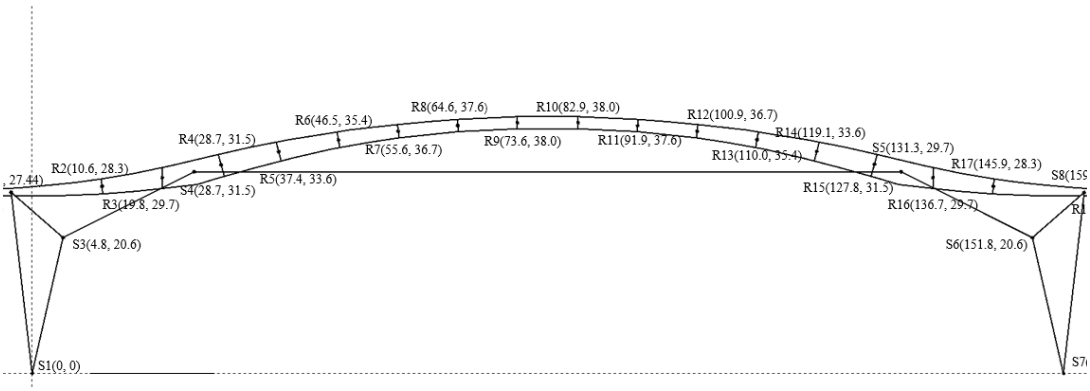
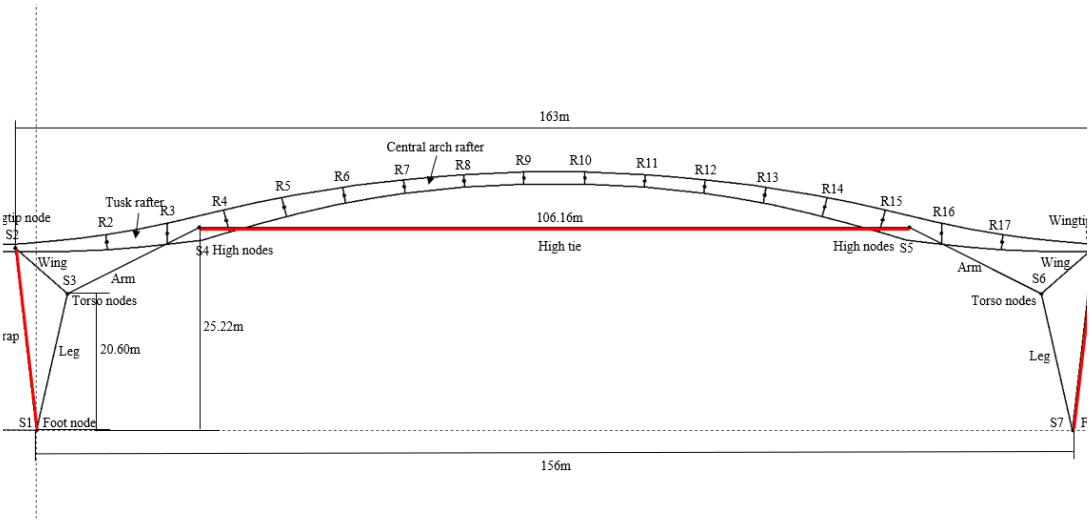
1 Deliverable 1: Geometry

The roof geometry was reconstructed from the supplementary drawing using a one-to-one SketchUp model and then transferred into GSA. As shown in Figure 1, points labelled R define the roof profile, while points labelled S define the support system and lower restraint points. The reconstructed model was checked against the original drawing on both sides of the roof and the overall geometric deviation was approximately 0.5%, which is acceptable for global FE modelling.

The second diagram shows the FE idealisation adopted for analysis. The curved roof was represented by short straight beam elements between adjacent roof nodes, while the high tie, straps, legs, arms and wings were represented as line members connected at the major nodes. This idealisation captures the main load path and preserves the key dimensions required for later loading, validation and mesh sensitivity studies.



(a) Reconstructed nodal geometry



(b) Idealised FE member layout

Figure 1: Geometry adopted for the global model.

2 Deliverable 2: Material, Section, and Loading Data

2.1 Material and Section Properties and Loading Data

The global FE model was defined using linear-elastic steel. In accordance with the brief, all members were assigned an elastic modulus of 200 GPa. A Poisson's ratio of 0.30 and density of 7850 kg m^{-3} were adopted so that self-weight could be included automatically in GSA. S355 steel was used for the primary roof members, which is consistent with the Heathrow T5 project information. The high tie was modelled with an initial prestress of 450 MPa.

Only vertical symmetric loading was considered. The characteristic permanent roof load was taken as the sum of the aluminium standing seam roof (0.10 kPa), structural deck (0.15 kPa), insulation/acoustic layers (0.40 kPa) and the service allowance (0.15 kPa), giving $G_k = 0.80$ kPa. The live load imposed on the roof was taken as $Q_k = 0.60$ kPa. Therefore, the governing vertical ultimate limit state (ULS) combination used in the model was

$$1.35G_k + 1.5Q_k = 1.35(0.80) + 1.5(0.60) = 1.98\,\text{kPa}.$$

Using a width of 36 m for one transverse frame, it is found that a line load of 71.28 kN m^{-1} is applied in addition to the self-weight of the primary members.

The strap members were modelled as rod elements. Therefore, only their axial stiffness was represented in the global model, using an equivalent steel area of $A = 2 \times 500 \times 40 = 40\,000\,\text{mm}^2$. The spacing between the two plates was neglected in this idealisation because the rod elements do not carry bending.

Table 1: Section data adopted for the global model

Member	Section	Modelling note
Arch rafter	Welded box, 800 mm wide, 50 mm wall thickness	Depth varies from about 1.8 m to 3.8 m along the roof profile
Tusk rafter	Welded I-section, 50 mm plate thickness	Overall depth varies from about 0.9 m to 3.8 m
Legs and arms	Circular Hollow Section (CHS) 914 mm × 40 mm	Taken directly from the supplementary sketch
Wings	CHS 711 mm × 50 mm	Taken directly from the supplementary sketch
Internal/external ties	CHS 457 mm × 40 mm	Thickness not stated in the source, so 40 mm was assumed
High tie	Twin locked-coil strands, 2 × 115 mm	Prestressed to 450 MPa
Straps	Equivalent rod area 40 000 mm^2	Based on two 500 mm × 40 mm plates; bending stiffness neglected

2.2 Assumptions

The assumptions made were the use of a linear-elastic S355 steel model, uniform wall thicknesses where indicated, a 36 m tributary width for each transverse frame, automatic inclusion of self-weight through the material density, and axial-only idealisation of the strap members.

3 Deliverable 3: Initial Model Validation

3.1 Assumptions

One important modelling assumption is that the real roof has been reduced from a 3D structural system to a 2D frame idealisation. This means only the in-plane behaviour of one transverse frame is represented directly, while out-of-plane bending, torsion and any load-sharing between adjacent frames are neglected. As a result, the model cannot capture twisting of the curved rafters, weak-axis bending caused by geometric eccentricity, or other local 3D effects associated with the actual connection arrangement. This simplification is acceptable for an initial global study because the loading considered here is vertical and symmetric, so the dominant response is expected to remain in-plane, but it should still be recognised as a source of modelling error when interpreting the validation results.

3.2 Model Validation

3.3 Model Verification

Because a converged answer to an incorrect model is still wrong, the following basic verification checks were successfully performed on the model to ensure that the solution process was correctly implemented:

- **Error Warnings:** The GSA solver completed the static analysis with no warnings regarding matrix singularities or instabilities.
- **Overall Equilibrium Check:** The sum of the vertical reaction forces (R_z) at the pinned supports was confirmed to equal the total applied vertical gravity loads ($\sum F = R \approx 0$).
- **Symmetry:** Because the structural geometry and the critical ULS load combination are perfectly symmetric, the resulting internal forces and deflected shape were verified to be entirely symmetric.
- **Qualitative Check:** The deflected shape aligns with physical expectations; the vertical loads cause the arch crown to displace downwards, putting the main ties into pure axial tension.

3.4 Model Adequacy

Taken together, the modelling assumptions, validation considerations and verification checks indicate that the global frame model is adequate for an initial study of the roof under symmetric vertical loading. However, it should still be understood as a simplified global model rather than a full representation of the three-dimensional structural behaviour.

4 Deliverable 4: Mesh Refinement Sensitivity Study

4.1 Introduction to Discretisation Error

Finite element analysis provides an approximation, not the exact analytical solution. When dividing a continuous structure into finite elements, discretisation error is introduced from two main sources: element behaviour error (dependent on the chosen shape functions) and mesh density error (dependent on element sizing). Mesh refinement evaluates whether this approximation is acceptable by increasing the degrees of freedom (DoF) until the results converge. To isolate the mesh density error without distorting the global physical behaviour, refinement was strictly limited to the continuous curve of the arch rafter. Based on St-Venant's principle, the detailed stress distribution near load applications or complex joints only matters locally. Refining the complex intersections at the tusk rafters, legs, and high ties could artificially induce stress singularities due to idealised geometry (such as re-entrant corners), which do not converge. Therefore, those boundary members remained constant while the arch was discretised into varying densities.

4.2 Convergence Study Methodology

Following standard FEA procedures, the convergence study was conducted as follows:

To evaluate convergence, it is necessary to check global quantities away from boundary singularities. Maximum vertical crown deflection (δ_z) and maximum arch bending moment (M_y) were selected. Peak local stresses were ignored as they are highly dependent on constraint application and may not converge.

Three mesh densities were evaluated to test the sensitivity of the global response to geometric discretisation:

- **Mesh 1 (Coarsest):** Highly simplified faceted geometry merging multiple coordinate segments.
- **Mesh 2 (Coarse):** Partially merged geometry.
- **Mesh 3 (Baseline):** The exact original coordinate nodes defined in the project data.

The results for the critical ULS vertical load combination ($1.35G_k + 1.5Q_k + 1.0P_s$) are tabulated below.

Table 2: Effect of mesh density and model size on key responses

Mesh	Arch elements	DoF	Deflection (mm)	Axial force (kN)	Bending moment (kNm)	% change vs Mesh 3
Mesh 1	14	53	54.53	21490	19250	21.3%
Mesh 2	16	59	45.44	20040	6316	1.1%
Mesh 3	22	77	44.94	19490	6023	0.0%

A solution is considered converged when further refinement causes negligible change in the target quantity. The following strict convergence criterion was applied between iterations:

$$\frac{|\delta_n - \delta_{n-1}|}{\delta_n} < 2\% \tag{1}$$

As demonstrated in Table 2, transitioning from the coarsest faceted model to the baseline geometry produced a noticeable change in both deflection and internal force, showing that the coarser meshes artificially stiffened the arch. Using Mesh 3 as the baseline, Mesh 1 differs by 21.3% in crown deflection whereas Mesh 2 differs by only 1.1%, indicating that the solution is approaching convergence.

4.3 Conclusion

Based on the achieved convergence, Mesh 3 is deemed structurally sufficient. It provides an accurate global approximation without the unnecessary computational expense of higher-order refinement, and has been adopted for the remainder of this report.

5 Deliverable 5: Influence of Prestressing Assumptions

5.1 Aim and Comparison Basis

The brief requires a comparison between the coursework simplification, in which high-tie prestress is applied after complete erection, and the construction process described in the supplementary documentation, where the centre arch was assembled and prestressed at low level before lifting and staged load transfer. A full staged-construction analysis was not undertaken here; instead, two simplified bounding models were compared, one with prestress neglected and one with the prescribed 450 MPa prestress included. This gives a practical indication of how sensitive the global response is to activation of the tie.

5.2 Quantitative Comparison

Table 3: Simplified sensitivity of global response to high-tie prestress

Response quantity	**Prestress neglected**	**Prestress included**	**Change with prestress**
Maximum vertical displacement (mm)	76.41	44.94	−41.2%
Maximum arch rafter bending moment (kNm)	26520	18270	−31.1%
Maximum axial force in arch rafter (kN)	13170	6023	−54.3%
High tie force (kN)	6736	11760	+74.6%
Horizontal reaction at supports (kN)	6430	5699	−11.4%

5.3 Discussion of Results

Table 3 shows that including prestress reduces maximum vertical displacement by 41.2%, arch rafter bending moment by 31.1%, axial force in the arch rafter by 54.3%, and horizontal reaction at the supports by 11.4%, while increasing high-tie force by 74.6%. The global force flow therefore becomes more arch-and-tie dominated and less bending-dominated once the tie is activated. Since the real roof was prestressed before final lifting and load transfer, its behaviour is expected to lie closer to the prestressed case, although the present models do not capture the locked-in effects of incremental stressing, creep adjustment and staged load transfer.

5.4 Implications for Design and Tolerance

The results show that the predicted design actions in the rafters, high tie and supports are sensitive to the prestress assumption, so neglecting prestress would overestimate deflection and arch bending while underestimating tie force. From a design perspective, this affects not only the sizing of the arch rafters themselves, but also the actions transmitted into the high-tie anchorage, support nodes and critical connection regions such as the rafter splice. A model that ignores prestress could therefore be conservative for some bending checks while being non-conservative for tie force and connection demand.

The supplementary paper also shows that tolerance was controlled through paired cable stressing, geometry monitoring, preset positioning, staged load transfer and final tension adjustment after creep. This indicates that final geometry and final force state are linked: deviations in fit-up, support position or closure of interfaces during erection can change how load is redistributed and therefore alter the locked-in forces present in the completed roof. For the remainder of the report, the prestressed model is therefore adopted as the minimum realistic representation of the roof, while recognising that explicit staged construction would be required to quantify locked-in stresses and erection tolerances more accurately. Overall, the comparison confirms that the real roof is much better represented by a prestressed global model than by an unprestressed frame model.

6 Deliverable 6: Boundary Conditions for More Detailed Analysis

For a more detailed FE study, the rafter splice between the tusk rafter and arch rafter should be extracted from the global frame model and analysed as a local submodel. The purpose of the local model would be to resolve the stress distribution in the splice plates, bolts or welds, and the adjacent box and I-section walls, which cannot be captured accurately by the beam idealisation used in the global model.

The local model should include a short length of member on each side of the splice so that the boundary conditions are applied away from the connection itself. This reduces artificial stress concentrations caused by restraining the model too close to the detail. A practical choice would be to include approximately one section depth of the arch rafter and one section depth of the tusk rafter beyond the splice on either side.

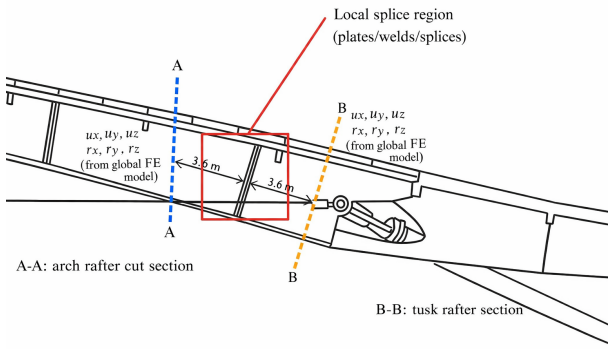


Figure 2: Recommended boundary conditions for the local splice model

The preferred boundary condition is to impose the displacement and rotation components extracted from the global beam model at the two cut faces of the local model. In practice, each cut face can be coupled to a reference node so that the sectional deformation remains compatible with the global model. The six kinematic quantities imposed at each end are the three translations (u_x, u_y, u_z) and the three rotations (r_x, r_y, r_z). This approach is preferable to simply fixing one end, because it reproduces the actual deformation state of the connection and avoids introducing artificial restraint moments.

The corresponding section forces should also be checked against the global model outputs. The main actions expected at the splice are axial force N, vertical shear V, bending moment M, and a smaller torsional component due to the curved roof geometry and eccentricity of the connection detail. If the software or modelling approach makes direct displacement transfer difficult, an acceptable alternative is to apply the equivalent set of sectional resultants at the two cut faces, provided that overall equilibrium is maintained.

To prevent rigid body motion in the local model, one reference node may be lightly stabilised after the global displacements have been assigned. This stabilisation should only remove numerical singularities and must not alter the transferred load path. Contact, bolt preload, or weld representation can then be added depending on the intended level of detail. For the purposes of this coursework, the key requirement is that the splice boundaries are driven by the global model response rather than by arbitrary fixed supports.

Section cuts are located approximately one local section depth away from the splice connection along the axis of the member. This avoids local stress disturbance. As the depth varies along the rafter, the local section depth is equal to the splice-region depth (3.6 m), cuts A–A and B–B are placed approximately 3.6 m from the splice on either side. At both cut faces, the translational and rotational displacements extracted from the global FE model are imposed.This ensures compatibility between the local splice model and the global structural response.